Autonomous Navigation, Path Planning, and Trajectory Control for a Bipedal Wheeled Robot

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***Abstract*—Bipedal wheeled robots offer a unique hybrid of the speed of wheeled locomotion and the spatial agility of legged bionics. However, they function as highly unstable wheeled inverted pendulums (WIP) requiring continuous, high-frequency dynamic stabilization. Traditional navigation architectures relying on centralized operating systems (e.g., ROS) introduce non-deterministic computational latency, making them unviable for real-time inverted pendulum control. This paper presents a strictly distributed embedded architecture for simultaneous autonomous navigation, path planning, and trajectory control. By offloading multi-sensor perception to a dedicated Sequential Kalman filter and operating a highly optimized, static-memory Hybrid A* path planner natively alongside a unified Linear Quadratic Regulator (LQR), the robot seamlessly executes complex evasive maneuvers and kinematically feasible trajectories without violating the strict physical equilibrium required to remain upright.**

***Index Terms*—Bipedal Robot, Wheeled Inverted Pendulum, Sensor Fusion, Hybrid A*, LQR Control, Autonomous Navigation**

I. Introduction

As autonomous mobile robots transition into unstructured human environments, the demand for physical agility and a minimal footprint has driven the development of bipedal wheeled robots. By modeling the chassis as a Wheeled Inverted Pendulum (WIP), these platforms achieve high-speed, energy-efficient locomotion. However, unlike traditional rovers, a bipedal wheeled robot is inherently unstable and requires continuous actuation simply to remain upright.

A primary challenge in this domain is the simultaneous execution of two conflicting computational tasks: maintaining the strict, low-latency control loop required to balance the pendulum, while processing dense spatial data for autonomous obstacle avoidance and path planning. Standard navigation architectures, such as Ubuntu running the Robot Operating System (ROS), rely on heavy multi-sensor fusion and complex graph-search algorithms. These centralized setups introduce non-deterministic computational latency; for a dynamically balancing robot, missing a 5-millisecond control loop deadline results in catastrophic mechanical failure.

To resolve these computational bottlenecks, this paper proposes a strictly distributed embedded hardware architecture. The workload is bifurcated: complex environmental perception is offloaded to a dedicated vision node running a custom sequential Kalman filter, while the critical balancing and trajectory generation are isolated on a highly deterministic microcontroller executing a static-memory Hybrid A* algorithm and an integrated Linear Quadratic Regulator (LQR) controller.

II. Related Work

The development of autonomous bipedal wheeled robots represents a convergence of dynamic control, kinematically aware path planning, and real-time perception.

A. Bipedal Mechanisms and Stabilization

Early heavy-duty wheel-legged platforms, such as Boston Dynamics' Handle, utilized hydraulic drives and centralized dynamic control centers. However, for safe indoor human interaction, the industry shifted toward lightweight, electrically actuated wheeled inverted pendulum (WIP) chassis [1]. Maintaining the balance of a WIP necessitates continuous high-frequency actuation. While Proportional-Integral-Derivative (PID) controllers are computationally simple, they are highly susceptible to saturation under complex loads. Consequently, optimal control strategies, specifically the Linear Quadratic Regulator (LQR), have become the standard for maintaining strict bipedal dynamics against sudden disturbances by continuously calculating optimal motor torques across multiple states [1].

B. Spatial Navigation and Kinematic Planning

Standard autonomous navigation heavily utilizes sampling-based planners like Rapidly-exploring Random Tree Star (RRT*) and discrete graph-search algorithms like conventional A*. However, RRT* implementations generate straight-line segments with sharp, kinematically unfeasible angular transitions that induce severe impulsive disturbance torques, destabilizing bipedal control loops [2]. Similarly, discrete grid-search algorithms inherently assume holonomic movement [3]. To bypass heavy mathematical post-processing, Hybrid A* has emerged as a state-of-the-art framework that operates in a continuous 3D state space (X, Y, θ). By utilizing specific motion primitives based on a vehicle's minimum turning radius, it natively resolves non-holonomic constraints.

C. Perception and Sensor Fusion

Robust environmental awareness typically relies on multi-sensor setups, with LiDAR often utilized for deep spatial perception [4]. Processing dense point clouds via algorithms like the Vector Field Histogram (VFH) requires substantial computational overhead, introducing prohibitive latency for balancing bipeds. Furthermore, reliance on a single sensor exposes a system to inherent hardware blind zones [5]. Cooperative sensor fusion based on the Kalman filter significantly improves real-time responsiveness and actively mitigates environmental interference [6]. This project builds upon these principles by

proposing a lightweight 1D sequential Kalman filter tailored specifically for inverted-pendulum physics, paired with robust planar odometry fusion [7].

III. Environmental Perception and Navigation

To safely execute planned trajectories without introducing latency into the stabilization loop, environmental perception was entirely offloaded to a dedicated vision node (ESP32-S3), utilizing an asymmetrical UART link to transmit clean spatial data to the main STM32 controller.

A. Cooperative Sensor Fusion

The hardware architecture relies on two HC-SR04 ultrasonic sensors for proximity and transparent-object detection, paired with an LD06 Time-of-Flight (ToF) LiDAR software-restricted to a forward 30-degree cone (-15° to $+15^\circ$). To completely bypass the computationally heavy matrix inversion typically required for sensor fusion, a novel 1D Sequential Kalman Filter was developed.

The fusion occurs in two stages. First, the dual ultrasonic sensors are fused to eliminate acoustic cross-talk and yield a unified acoustic measurement zU_{fused}. Second, this unified reading is fused with the optical laser data z_L to yield the final reliable distance value.

B. Dynamic Trust Logic

Because the sequential architecture isolates the measurement covariances (R_U for acoustics and R_L for optics), the system utilizes Dynamic Trust Logic to actively mitigate edge-case sensor failures :

- **Elevation Override:** If the ultrasonics detect an object at close range while the LiDAR reports clear space (indicating an obstacle below the 2D laser plane), R_L is artificially maximized. Trust shifts entirely to the ultrasonic data.
- **Acoustic Ghosts:** If sound waves deflect off a smooth, angled surface, causing the ultrasonics to report a false clear path while the LiDAR detects an object, R_U is maximized to trust the LiDAR 100%.
- **Panic Stop:** For objects at point-blank range (< 30 cm), where ToF LiDARs experience hardware blind zones, trust reverts heavily to the acoustic fail-safe.

IV. Embedded Kinematic Path Planning

The path planning subsystem was designed to resolve the tension between the robot's non-holonomic kinematic constraints and real-time execution limits. Initial simulations utilizing standard RRT* generated kinematically unfeasible angular transitions, and post-processing via B-Splines introduced excessive computational overhead.

A. Static Memory Architecture

To guarantee deterministic execution times and eliminate heap fragmentation caused by dynamic memory allocation, the core algorithm was transitioned to a Hybrid A* planner operating on a continuous $SE(2)$ state space. The algorithm was optimized into a portable C99 library. By locking memory to a statically allocated node pool and a 3D discretized grid for visited-state lookups, the total memory footprint was fixed at 315 KB, allowing it to run natively on the STM32 controller.

B. The Four-Stage Execution Pipeline

The trajectory generation executes in four sequential stages:

1) **Kinematic Map Inflation:** Artificial expansion of obstacles by the robot's radius plus a safety margin.
2) **Hybrid A* Expansion:** Searching the grid using motion primitives restricted by the biped's minimum turning radius, natively generating executable headings.
3) **Iterative Gradient Descent Smoothing:** Waypoints are pulled toward the midpoint of neighbors (Laplacian smoothing). "Launch and Approach Gates" force straight-line departures and arrivals.
4) **Anti-Snap Trajectory Filtering:** Yaw-rate clamping guarantees that the rate of change of heading ($\Delta\theta$) never exceeds the physical capabilities of the inverted pendulum control loop.

C. Replanning State Machine

To avoid computationally expensive replanning triggered by transient obstacles, a three-state executive machine was implemented: `STATE_MOVING`, `STATE_WAITING`, and `STATE_REPLANNING`. Upon detecting a sudden obstacle, the robot enters the waiting state for a 2.0-second patience window. If the obstacle clears, execution resumes immediately; otherwise, a double-buffered bypass calculation is triggered.

V. Trajectory Tracking and Control

The control architecture models the robot as a WIP, where the same wheel torques must maintain the upright posture and follow the desired spatial path.

A. Planar Error Kinematics

The controlled planar pose is defined as $\mathbf{p} = [x, y, \theta]^T$. The trajectory tracking error is expressed relative to the desired heading direction:

$$e_x = \cos(\theta_d)\Delta x + \sin(\theta_d)\Delta y \tag{1}$$

$$e_y = -\sin(\theta_d)\Delta x + \cos(\theta_d)\Delta y \tag{2}$$

Where e_x represents error along the path, and e_y represents lateral deviation. The trajectory tracking error vector is defined as $\mathbf{e}_{traj} = [e_x, e_y, e_\theta]^T$.

B. Integrated Control

Instead of treating stabilization and navigation as two isolated problems, the system employs a Combined LQR block. The locomotion state vector includes the body tilt φ, traveled distance s, heading θ, tilt rate $\dot{\varphi}$, forward velocity v, and yaw rate ω.

The unified feedback error vector is defined as:

$$\mathbf{e} = \begin{bmatrix} \varphi \\ s - s_d \\ e_\theta \\ \dot{\varphi} \\ v - v_d \\ \omega - \omega_d \end{bmatrix} \tag{3}$$

The desired body tilt and tilt rate are strictly set to zero to preserve upright equilibrium. The LQR control law simultaneously calculates the required optimal wheel torques:

$$\mathbf{u}_{des} = -\mathbf{K}_{LQR}\mathbf{e} \tag{4}$$

VI. Experimental Results

The system's performance was evaluated through a combination of Software-in-the-Loop (SIL) simulations and hardware validation to isolate and verify each subsystem before full integration.

A. Sensor Fusion and Telemetry Performance

Initial sensor baselines confirmed that while the HC-SR04 ultrasonic array reliably detected flat objects up to 2.5 meters, it was highly susceptible to acoustic deflection on angled surfaces. Conversely, the LD06 LiDAR provided millimeter-precise spatial data but suffered from hardware blind zones at point-blank range (< 30 cm). The custom 1D Sequential Kalman Filter successfully merged these modalities. Physical testing validated the Dynamic Trust Logic: during an "Elevation Override" scenario (a low-lying obstacle below the 2D laser plane), the system successfully maximized the optical noise covariance (R_L), snapping trust entirely to the ultrasonic array.

Furthermore, resolving communication latency was critical. Initial testing of wireless telemetry (Single-Core Wi-Fi and Distributed Wireless via dual ESP32s) resulted in severe packet drops and non-deterministic micro-stalls that violated the 5ms control loop. Consolidating the pipeline to a hardwired, asymmetrical UART architecture—utilizing a 921,600 baud Direct Memory Access (DMA) link to the STM32—completely eliminated these bottlenecks, ensuring a continuous, jitter-free spatial data stream.

B. Path Planning and SIL Validation

The embedded C99 Hybrid A* algorithm was rigorously tested against a standard MATLAB RRT* baseline. While RRT* produced jagged, kinematically unfeasible trajectories requiring heavy B-Spline post-processing, the Hybrid A* planner natively generated continuous, smooth paths bounded by the robot's physical constraints. By utilizing a strictly static memory architecture (capping the node pool and 3D closed-list array), the algorithm achieved an $O(1)$ memory footprint of exactly 315 KB, safely within the STM32's SRAM budget.

The Replanning State Machine was validated via a comprehensive 52.05-second SIL mission comprising 1,041 telemetry frames at 20 Hz. At $t = 8.05$ s, a mock obstacle was injected at 1.2 m, breaching the 1.5 m safety threshold. The system immediately halted trajectory execution and transitioned to `STATE_WAITING`. The obstacle was cleared at $t = 9.10$ s. Because this 1.05-second delay fell well within the 2.0-second patience window, the system seamlessly resumed `STATE_MOVING` without wasting CPU cycles on a map inflation and bypass calculation. Crucially, the trajectory logic mathematically guaranteed kinematic safety; with a speed of 0.2 m/s and a minimum turning radius of 0.5 m, all heading changes were strictly clamped to a maximum of 0.02 rad/frame.

C. Stabilization and Integrated Control

The LQR controller was first evaluated strictly for stabilization across three distinct mechanical height configurations (h_{min}, h_{mid}, h_{max}). Subjected to an initial tilt disturbance of $\varphi_0 = 0.1$ rad (approximately 5.7°), the controller successfully rejected the disturbance and returned the chassis to an upright equilibrium within 4 seconds for all stances, without inducing unstable oscillations or yaw drift.

Finally, the fully integrated controller was tested, simultaneously managing planar trajectory tracking and vertical stabilization. Experimental data confirmed that the actual linear and angular velocities smoothly tracked the desired reference profiles. The forward error (e_x), lateral error (e_y), and heading error (e_θ) remained strictly bounded. Most importantly, the body tilt angle (φ) remained stable throughout the execution of complex non-holonomic steering maneuvers. This confirmed that the dynamically-aware paths generated by the Hybrid A* algorithm did not introduce destabilizing impulsive torques, proving the viability of the unified control architecture.

VII. Conclusion

This architecture successfully demonstrates that dense spatial navigation and dynamic inverted-pendulum stabilization can be achieved strictly on embedded hardware. By offloading environmental perception to a dynamically weighted sequential filter, and constraining path generation to an $O(1)$ static memory Hybrid A* planner, the system guarantees the execution of fluid, kinematically feasible trajectories. The unified LQR controller seamlessly regulates both path tracking and body stabilization, resulting in a highly agile, autonomous bipedal wheeled robot capable of operating safely in unstructured environments.

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